Stereo Depth Estimation

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Figures adapted from CIS 580, Kostas Daniilidis

Announcement

- HW1 due today at 11:59 pm
- 5 free late days, but need to email the TAs before due time

Recap: what problems?

$$\mu_i \, \boldsymbol{q}_i = \mathbf{T} + \lambda_i \, \mathbf{R} \boldsymbol{p}_i$$

Known values	Problem	To solve
$\{\mu_i, \lambda_i, \boldsymbol{p}_i, \boldsymbol{q}_i\}$	Procrustes / Kabsch	$(\mathbf{R}, \mathbf{T})$
$\{\lambda_i, \boldsymbol{p}_i, \boldsymbol{q}_i\}$	PnP (3D–2D bearings)	$(\mathbf{R}, \mathbf{T})$ $(+\{\mu_i\})$
$\{\boldsymbol{p}_i, \boldsymbol{q}_i\}$	two-view SfM (estimate $\mathbf{E}$)	$\mathbf{E} \Rightarrow (\mathbf{R}, \pm\hat{\mathbf{T}}),\ \{\lambda_i, \mu_i\}$

Last column of U is a translation

1) The last column of $\mathbf{U}$ spans $\mathrm{Null}(\mathbf{E}^\top)$

Transpose the SVD:

$$\mathbf{E}^\top = \mathbf{V}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{U}^\top.$$

If $\mathbf{E}^\top \boldsymbol{x} = \mathbf{0}$ then

$$\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{U}^\top \boldsymbol{x} = \mathbf{0} \;\Rightarrow\; \mathbf{U}^\top \boldsymbol{x} \in \mathrm{span}(\boldsymbol{e}_3) \;\Rightarrow\; \boldsymbol{x} \in \mathrm{span}(\boldsymbol{u}_3).$$

2) $\mathbf{T} \in \mathrm{Null}(\mathbf{E}^\top)$

Using $\mathbf{E} = [\mathbf{T}]_\times \mathbf{R}$,

$$\mathbf{E}^\top \mathbf{T} = \mathbf{R}^\top [\mathbf{T}]_\times^\top \mathbf{T} = -\mathbf{R}^\top ([\mathbf{T}]_\times \mathbf{T}) = -\mathbf{R}^\top (\mathbf{T} \times \mathbf{T}) = \mathbf{0}.$$

Rotations

Let $\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top$.
Define

$$\mathbf{W} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

candidate rotations are

$$\mathbf{R}_1 = \mathbf{U}\mathbf{W}\mathbf{V}^\top, \qquad \mathbf{R}_2 = \mathbf{U}\mathbf{W}^\top\mathbf{V}^\top.$$

Important detail

Ensure $\det(\mathbf{R}) = +1$; if $\det(\mathbf{U}\mathbf{V}^\top) = -1$, flip the sign of one column of $\mathbf{U}$ or $\mathbf{V}$.

How many solutions?

From decomposition you typically get **four** candidates:

$$(\mathbf{R}_1,\ +\hat{\mathbf{T}}),\ (\mathbf{R}_1,\ -\hat{\mathbf{T}}),\ (\mathbf{R}_2,\ +\hat{\mathbf{T}}),\ (\mathbf{R}_2,\ -\hat{\mathbf{T}}).$$

Often described as two rotation choices (a *twisted pair*) and an unknown translation sign.

Cheirality test

Given a candidate $(\mathbf{R}, \mathbf{T})$:

1. Triangulate a few correspondences to get 3D points $\boldsymbol{X}$.
2. Check depths are positive in both cameras:

$$z_1 > 0, \qquad z_2 > 0.$$

Pick the candidate where the largest number of points satisfy positive depth.

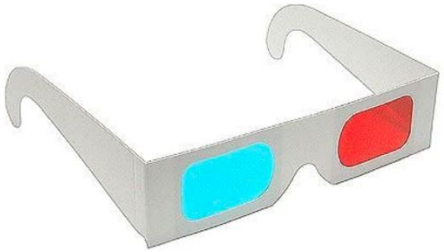
Stereo Depth Estimation

Stereogram

The sense of depth comes from the difference in what we see with two eyes

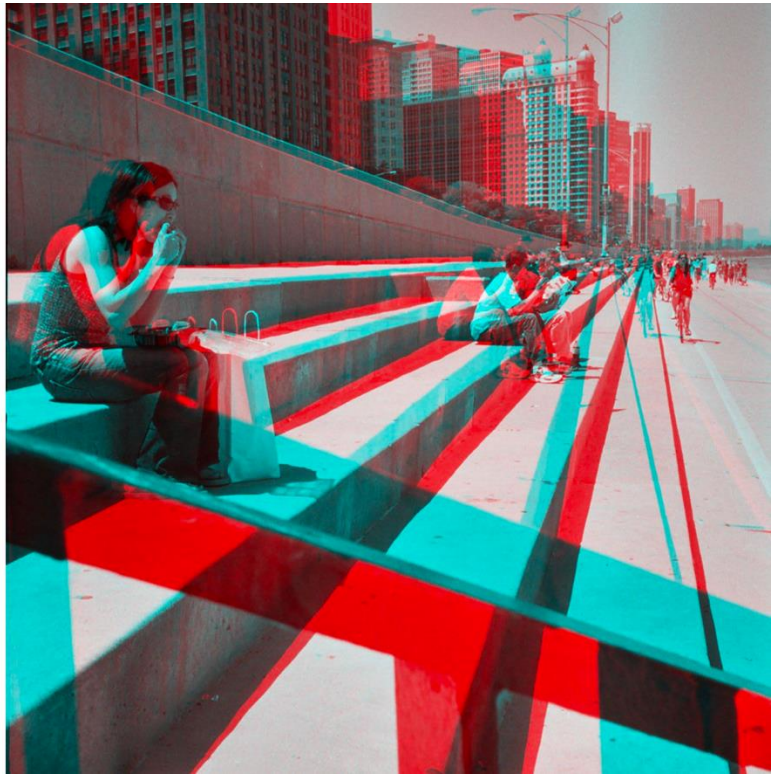
Calibrated two views

Anaglyphs and 3D movies



The left eye image is filtered to remove blue & green. The right eye image is filtered to remove red.

Hard to do this well for very bright colored scenes, but other approaches, e.g. polarization still work.

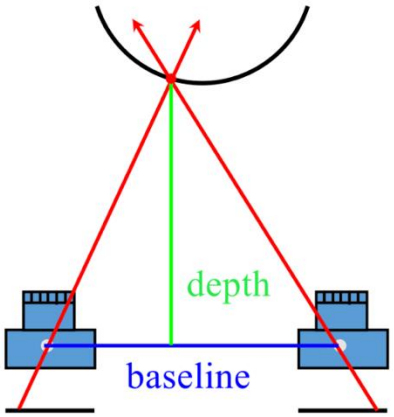


Stereo Cameras

Canon VR Lens

RealSense Stereo Camera

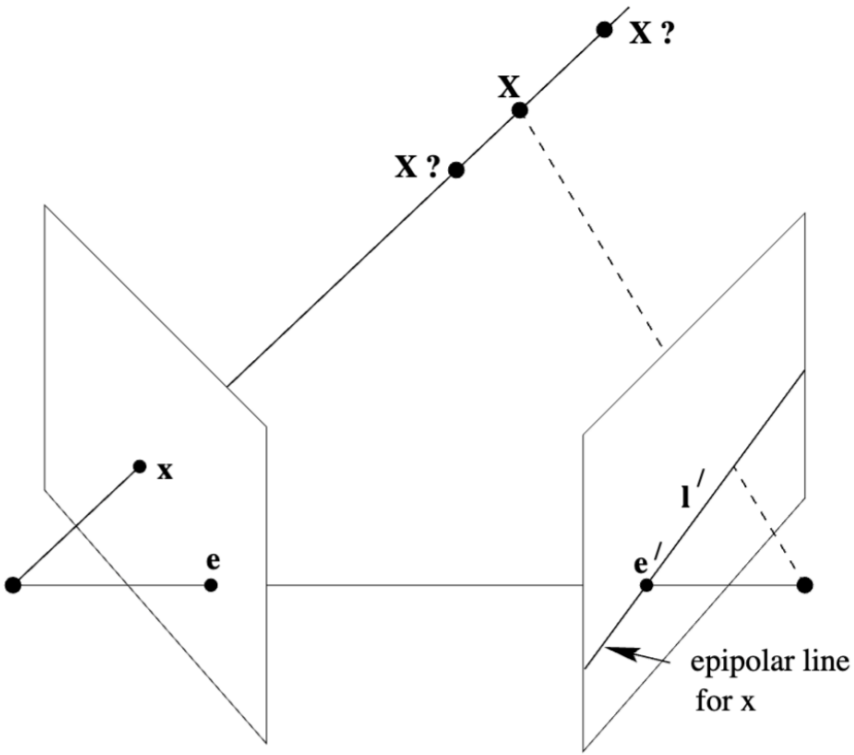
Basic Idea



Triangulate on two images of the same point to recover depth.

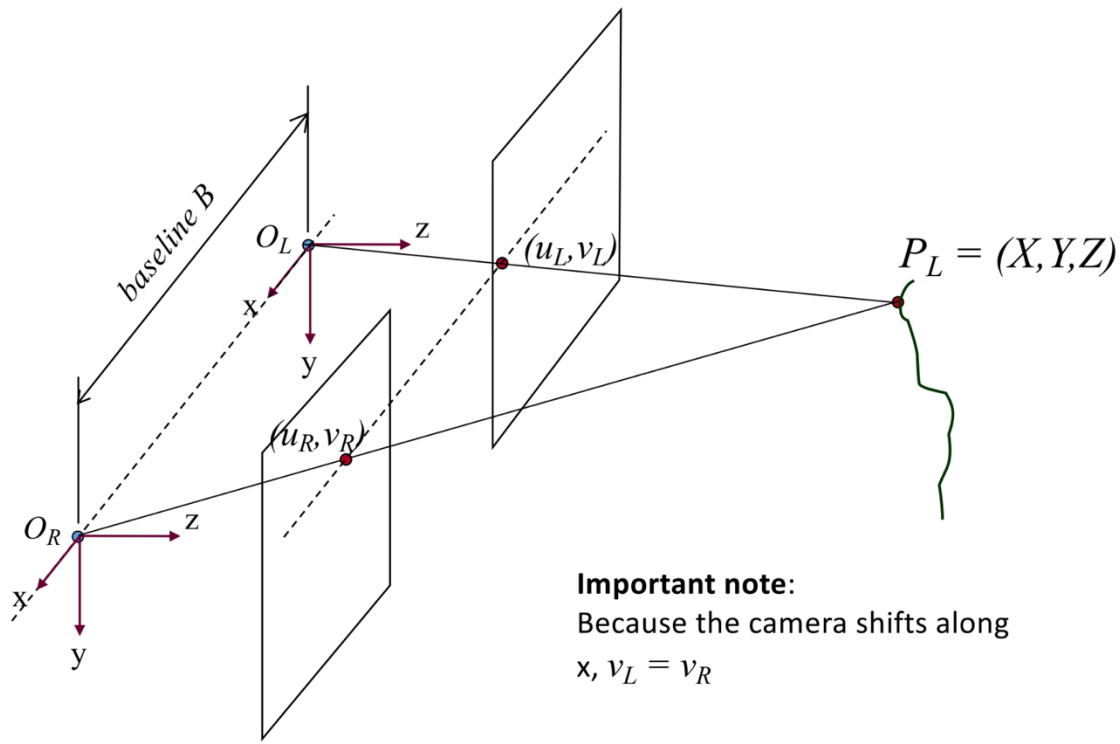
- Feature matching across views
- Calibrated cameras

Where are we searching?



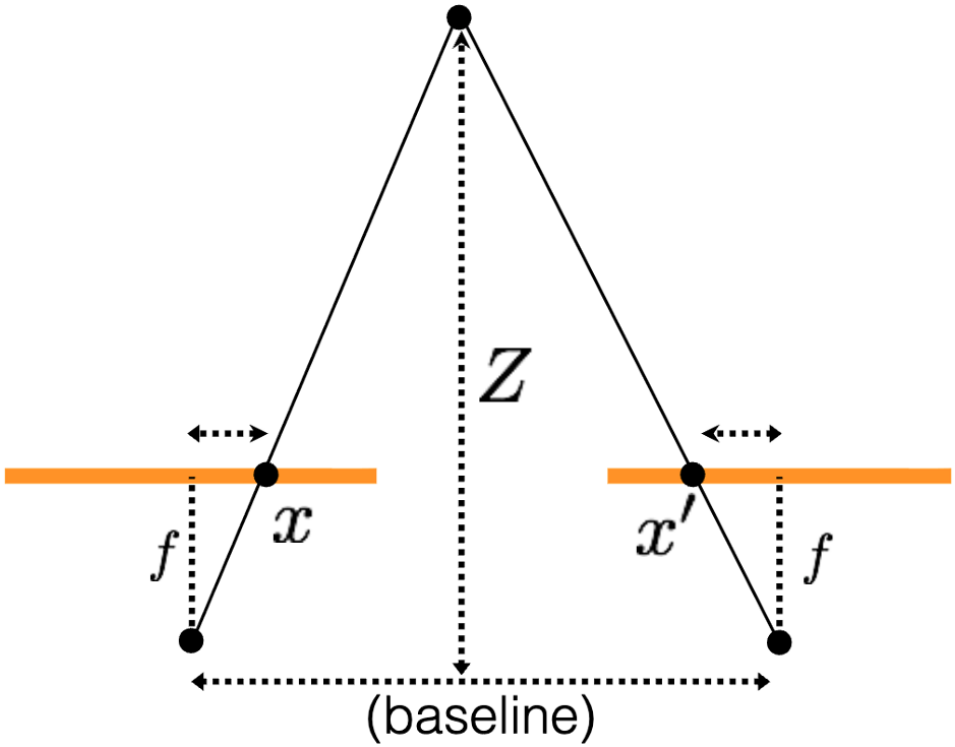
Triangulation, but in a more convenient way.

Stereo setup: parallel image planes

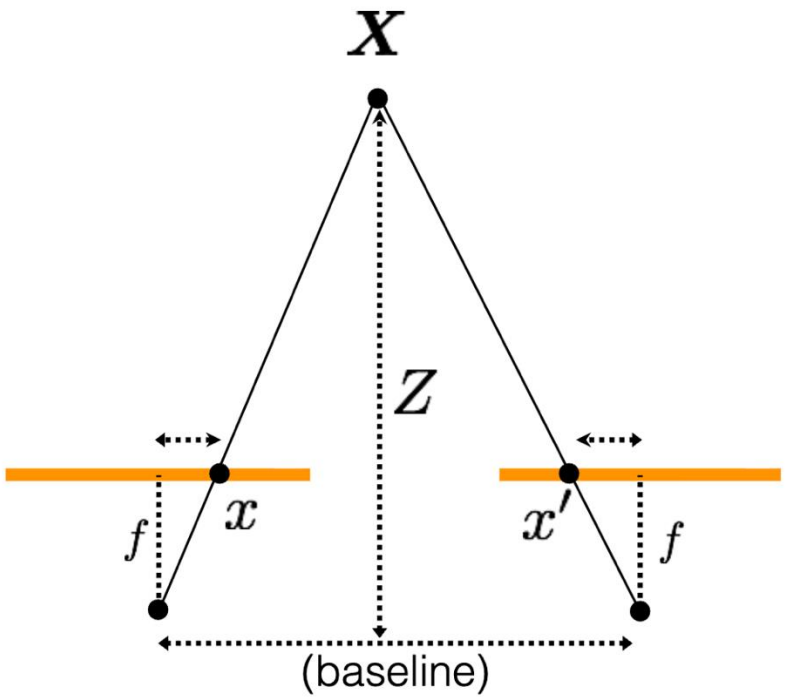


Important note:
Because the camera shifts along x, $v_L = v_R$

Disparity

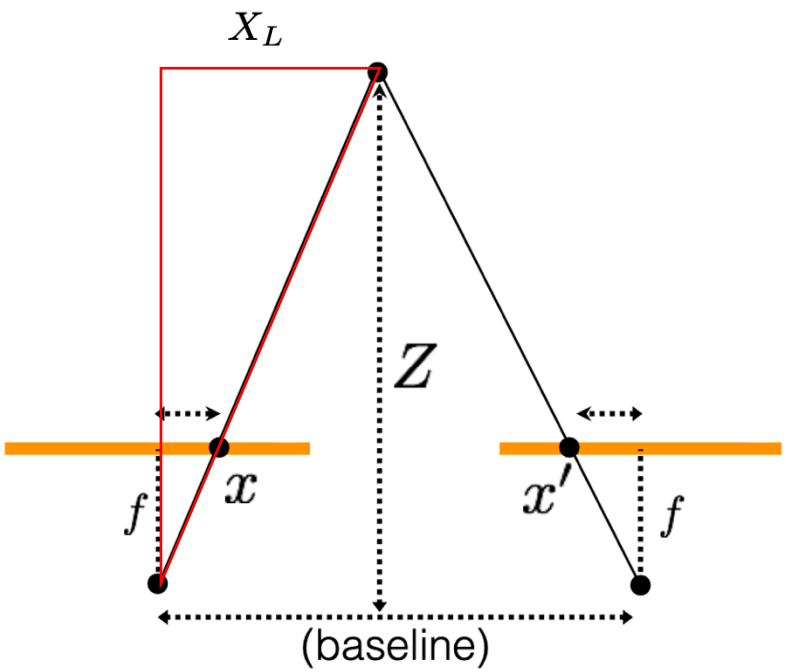


Depth from disparity



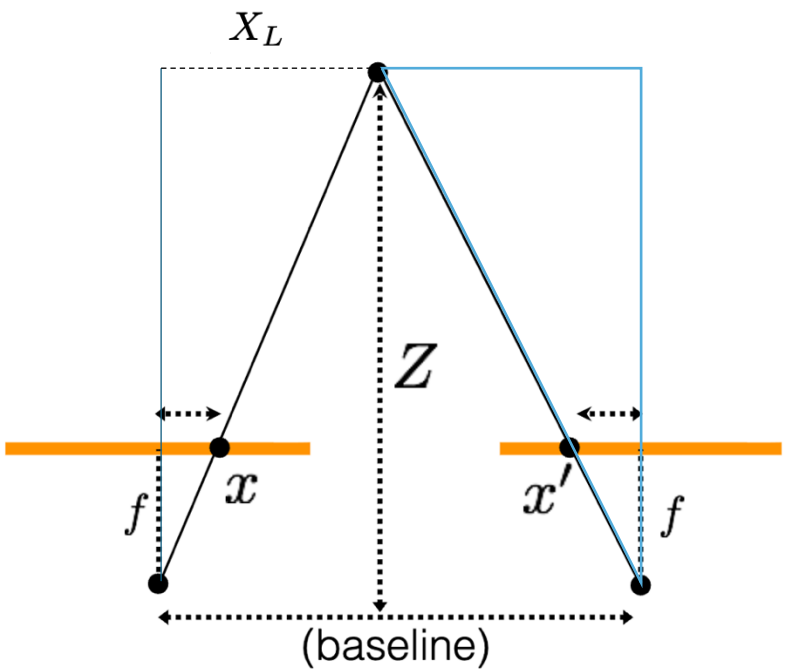
How do we relate Z (depth) with x, x' and f?

Depth from disparity



$$\frac{x}{f} = \frac{X_L}{Z}$$

Depth from disparity

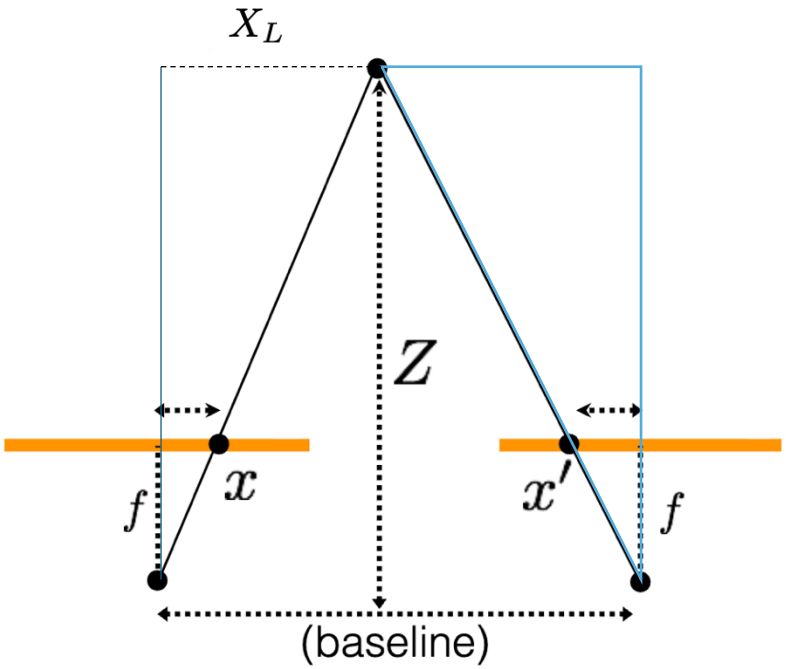


$$\frac{x'}{f} = \frac{X_R}{Z}.$$

Because the cameras are separated by baseline b along X,

$$b = X_L - X_R.$$

Depth from disparity



$$\underbrace{d}_{\text{disparity}} \triangleq x - x' = \frac{fX_L}{Z} - \frac{fX_R}{Z} = \frac{f(X_L - X_R)}{Z} = \frac{fB}{Z}.$$

Rearranging yields the classic depth-from-disparity relation:

$$\boxed{Z = \frac{fb}{d}}$$

Disparity

In rectified stereo, define disparity d as the horizontal shift:

$$d(u, v) \triangleq u_L - u_R.$$

- Large disparity $\Rightarrow$ close objects
- Small disparity $\Rightarrow$ far objects
- Disparity is typically measured in pixels (possibly subpixel)

Disparity plane discretization

$$Z = \frac{fb}{d}, \qquad d = \frac{fb}{Z}.$$

Differentiate:

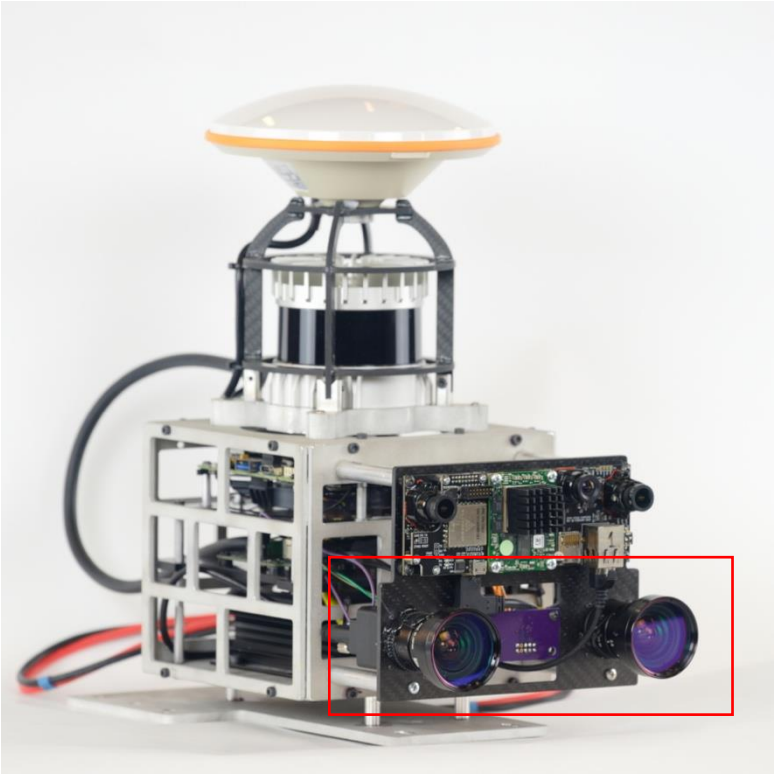
$$\frac{\partial Z}{\partial d} = -\frac{fb}{d^2}.$$

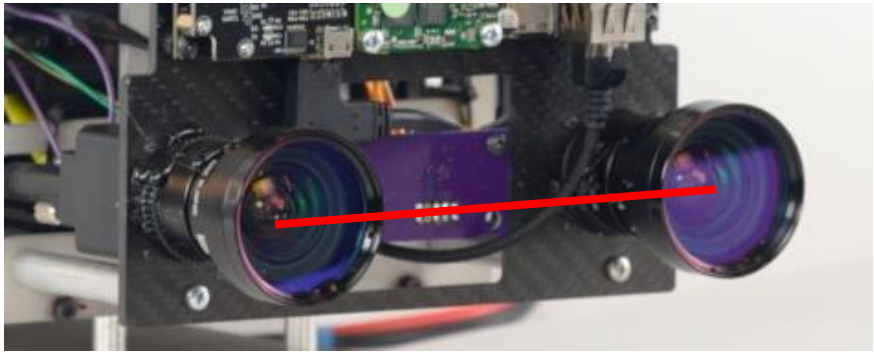
So (in magnitude) for a small disparity error δd,

$$|\delta Z| \approx \frac{fb}{d^2}\,|\delta d|.$$

If our disparity is only on whole pixels, the error can propagate to the final depth

Baseline





Baseline

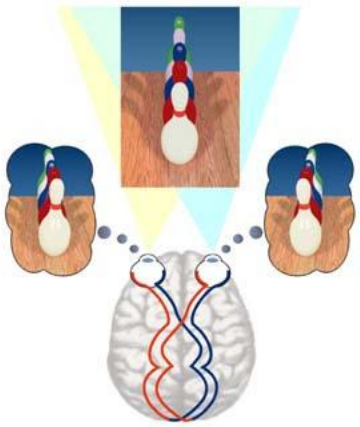
Stereo becomes harder with distance

Take a human-like baseline $b \approx 0.065\,\text{m}$. For a rectified camera model with focal length $f \approx 1000\,\text{px}$:

$$d \approx \frac{fb}{Z} = \frac{65}{Z\,[\text{m}]}\ \text{px}.$$

Example disparities:

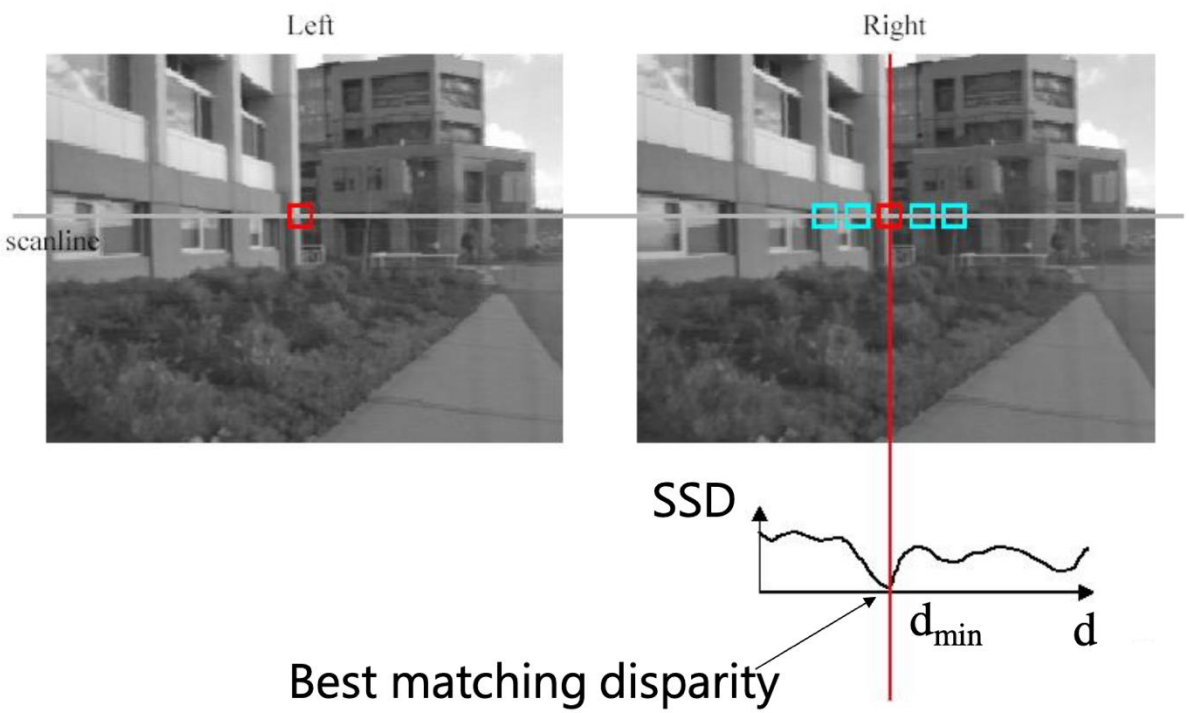
$$Z = 5\,\text{m}:\ d \approx 13\,\text{px} \qquad Z = 20\,\text{m}:\ d \approx 3.25\,\text{px} \qquad Z = 100\,\text{m}:\ d \approx 0.65\,\text{px}.$$



Humans lose stereo depth perception beyond ~20 meters

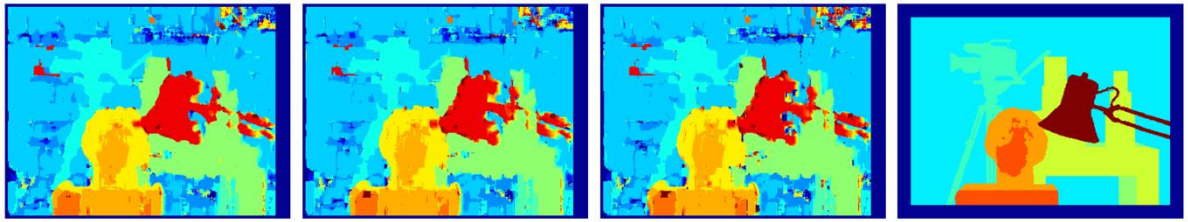
Matching

Stereo Block Matching



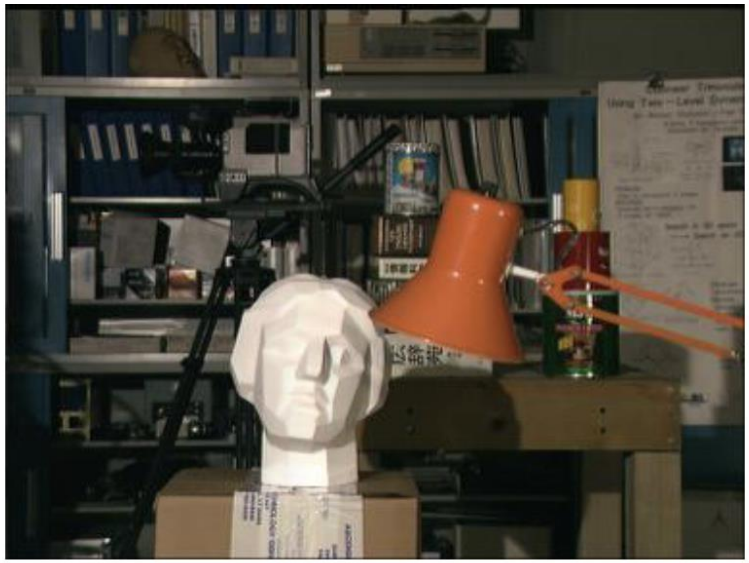
Matching Similarity/cost

Cost/Similarity	Rectified stereo matching
SAD	$\sum_{(i,j)\in W} \lvert I_L(x+i,\, y+j) - I_R(x-d+i,\, y+j) \rvert$
SSD	$\sum_{(i,j)\in W} \left(I_L(x+i,\, y+j) - I_R(x-d+i,\, y+j) \right)^2$
NCC	$\dfrac{\sum_{(i,j)\in W} I_L(x+i,\, y+j)\, I_R(x-d+i,\, y+j)}{\sqrt{\left(\sum_{(i,j)\in W} I_L(x+i,\, y+j)^2\right)\left(\sum_{(i,j)\in W} I_R(x-d+i,\, y+j)^2\right)}}$

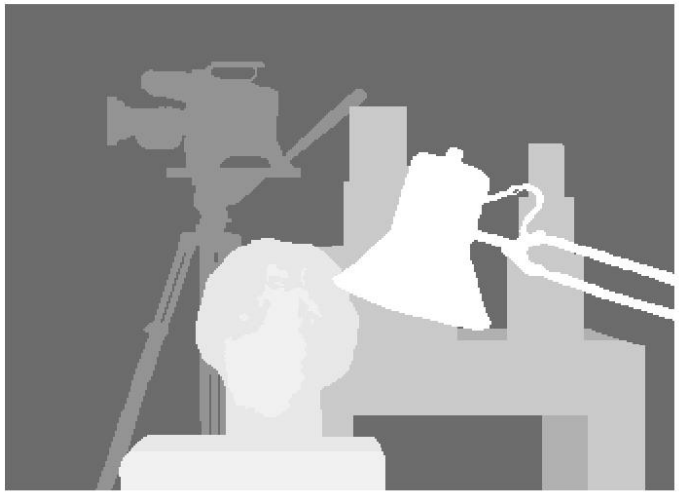


SAD SSD NCC Ground truth

Stereo Results



Scene



Ground truth

Cost volume

Compute costs for a disparity range $d \in \{0, \ldots, D-1\}$:

$$\text{Cost volume } C(u, v, d) \in \mathbb{R}^{H \times W \times D}.$$

- Stereo becomes: choose $d(u, v)$ that best explains each pixel

Aggregate costs over a support region $\Omega(p)$:

$$\bar{C}(p, d) = \sum_{q \in \Omega(p)} w(p, q)\, C(q, d).$$

Average over a region

After aggregation, choose the best disparity per pixel: Winner takes all

$$d^*(p) = \arg\min_d \bar{C}(p, d).$$

Left-right consistency

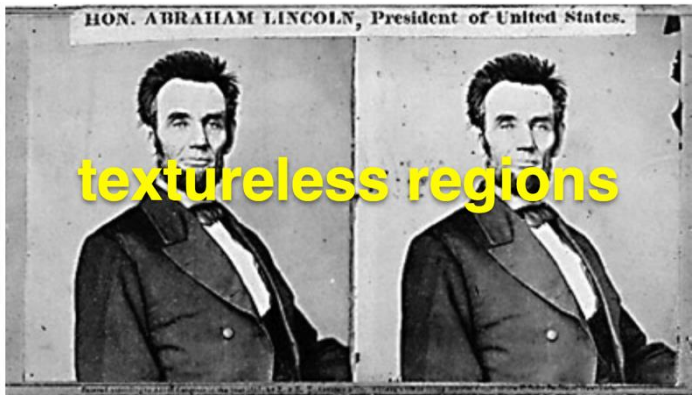
Compute disparity in both directions: $d_L(u, v)$ and $d_R(u, v)$.
Consistency test:

$$d_L(u, v) \approx d_R(u - d_L(u, v), v).$$

Occluded points create inconsistency

Occlusion: points that appear in one camera but not the other

Ambiguity: repeated pattern



Non-Lambertian surface

Stereo as energy minimization

Data term encourages matching

$$E(d) = \boxed{\sum_{p} C(p, d(p))} + \boxed{\lambda \sum_{(p,q)\in\mathcal{N}} \rho\big(d(p) - d(q)\big).}$$

Regularization term encourages smoothness

Assumption: Depth should change smoothly